

Algebra II with Trigonometry

Chapter 7.4 & 7.5

Olympiad Solutions

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[4626701_alg-2trig-h-chp-7-4-7-5-note-docx.pdf](#)

Solutions

1. If $\sin(2\theta) + \sin(\theta) = 0$, find all θ in $[0, 2\pi)$. Without computation, identify the symmetry among the solutions.

Solution:

First, use the double angle identity:

$$\begin{aligned}\sin(2\theta) + \sin \theta = 0 &\implies 2 \sin \theta \cos \theta + \sin \theta = 0 \\ \sin \theta (2 \cos \theta + 1) &= 0\end{aligned}$$

So, either $\sin \theta = 0$ or $2 \cos \theta + 1 = 0$.

Case 1: $\sin \theta = 0 \implies \theta = 0, \pi$.

Case 2: $2 \cos \theta + 1 = 0 \implies \cos \theta = -\frac{1}{2} \implies \theta = \frac{2\pi}{3}, \frac{4\pi}{3}$.

Symmetry: The solutions are symmetric about π (the unit circle center):

$$0 \leftrightarrow \pi, \frac{2\pi}{3} \leftrightarrow \frac{4\pi}{3}.$$

Thus, solutions:

$$\theta = 0, \pi, \frac{2\pi}{3}, \frac{4\pi}{3}$$

2. Find all real x for which $\cos^4 x + \sin^4 x = \frac{5}{8}$. What notable values of x satisfy this equation?

Solution:

We rewrite using the identity

$$\cos^4 x + \sin^4 x = (\cos^2 x + \sin^2 x)^2 - 2 \sin^2 x \cos^2 x = 1 - 2 \sin^2 x \cos^2 x:$$

$$\cos^4 x + \sin^4 x = 1 - 2 \sin^2 x \cos^2 x$$

$$1 - 2 \sin^2 x \cos^2 x = \frac{5}{8}$$

$$2 \sin^2 x \cos^2 x = 1 - \frac{5}{8} = \frac{3}{8}$$

$$\sin^2 x \cos^2 x = \frac{3}{16}$$

But $\sin^2 x \cos^2 x = \frac{1}{4} \sin^2 2x$:

$$\frac{1}{4} \sin^2 2x = \frac{3}{16} \implies \sin^2 2x = \frac{3}{4}$$

So

$$\sin 2x = \pm \frac{\sqrt{3}}{2}$$

Therefore

$$2x = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3} \pmod{2\pi}$$

So

$$x = \frac{\pi}{6}, \frac{\pi}{3}, \frac{2\pi}{3}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{4\pi}{3}, \frac{5\pi}{3}, \frac{11\pi}{6}$$

But only values in $[0, 2\pi)$.

Notable values: $x = \frac{\pi}{6}, \frac{5\pi}{6}$ (angles where $\sin 2x = \frac{\sqrt{3}}{2}$), $x = \frac{7\pi}{6}, \frac{11\pi}{6}$ (where $\sin 2x = -\frac{\sqrt{3}}{2}$).

$$x \in \left\{ \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6} \right\}$$

3. The equation $\tan 3\theta = \tan \theta$ holds for certain θ in $[0, 2\pi)$. How many distinct solutions does it have in this interval? (Do not solve explicitly; analyze using periodicity and symmetry.)

Solution:

The equation $\tan 3\theta = \tan \theta$ holds when

$$3\theta = \theta + k\pi \quad (k \in \mathbb{Z}) \implies 2\theta = k\pi \implies \theta = \frac{k\pi}{2}$$

Each θ in $[0, 2\pi)$ for $k = 0, 1, 2, 3$ yields $\theta = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$.

But this is not all, because $\tan 3\theta = \tan \theta$ also when both tangent arguments differ by integer multiples of π :

$$3\theta = \theta + m\pi, \implies \theta = \frac{m\pi}{2}$$

But for all $m = 0, 1, 2, 3$ ($0 \leq \frac{m\pi}{2} < 2\pi$), gives 4 values.

But note: Some extra solutions could arise when both $\tan 3\theta$ and $\tan \theta$ are undefined (i.e., $\theta = \frac{\pi}{2}, \frac{3\pi}{2}$), but those are already counted above.

Therefore, the **number of distinct solutions in $[0, 2\pi)$ is 4.**

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4. Let $\sin 5\theta - \sin 3\theta = \cos 4\theta$. Using sum-to-product identities, deduce for which θ in $[0, 2\pi)$ the equation is satisfied.

Solution:

Sum-to-product:

$$\sin 5\theta - \sin 3\theta = 2 \cos \left(\frac{5\theta + 3\theta}{2} \right) \sin \left(\frac{5\theta - 3\theta}{2} \right) = 2 \cos(4\theta) \sin(\theta)$$

So the equation becomes:

$$2 \cos(4\theta) \sin(\theta) = \cos(4\theta)$$

$$\cos(4\theta)[2 \sin \theta - 1] = 0$$

So, either

- $\cos 4\theta = 0$ or
- $2 \sin \theta - 1 = 0 \implies \sin \theta = \frac{1}{2}$

$$\text{Case 1: } \cos 4\theta = 0 \implies 4\theta = \frac{\pi}{2} + k\pi \implies \theta = \frac{\pi}{8} + \frac{k\pi}{4}$$

$$\text{For } \theta \in [0, 2\pi), k = 0, 1, \dots, 7: \theta = \frac{\pi}{8}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{7\pi}{8}, \frac{9\pi}{8}, \frac{11\pi}{8}, \frac{13\pi}{8}, \frac{15\pi}{8}.$$

$$\text{Case 2: } \sin \theta = \frac{1}{2} \implies \theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

Collect all values:

$$\theta = \frac{\pi}{8}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{7\pi}{8}, \frac{9\pi}{8}, \frac{11\pi}{8}, \frac{13\pi}{8}, \frac{15\pi}{8}, \frac{\pi}{6}, \frac{5\pi}{6}$$

5. For which real x does $\sin x = x$? Estimate to two decimal places using only logical reasoning and bounds.

Solution:

- For $x = 0$, $\sin 0 = 0$. So $x = 0$ is a solution.
- For $x > 0$, $\sin x < x$ for $x > 0$ (since tangent line is shallower than sine curve for $x > 0$), so no other positive solutions.
- For $x < 0$, $\sin x > x$ for small $x < 0$, but as $x \rightarrow -\infty$, $\sin x$ is bounded ($[-1, 1]$), while $x \rightarrow -\infty$, so the graphs cross somewhere for some negative x .

Find approximate negative x :

Test $x = -1$: $\sin(-1) \approx -0.84 > -1$ Test $x = -2$:

$\sin(-2) \approx -0.91 > -2$ Test $x = -3$: $\sin(-3) \approx -0.14 > -3$ Test $x = -4$: $\sin(-4) \approx 0.76 > -4$

But as x decreases, x becomes more negative, but $\sin x$ oscillates between -1 and 1 .

But is it possible that $\sin x = x$ for $x < 0$ except $x = 0$? Drawn graphically, the only intersection is at $x = 0$.

Thus,

$$\boxed{x = 0}$$

is the only real solution.

6. Find all θ in $[0, 2\pi)$ such that $4 \cos \theta \sin \theta + 3 \cos \theta = 0$. Factor and interpret geometrically to identify the solutions quickly.

Solution:

Factor $\cos \theta$:

$$4 \cos \theta \sin \theta + 3 \cos \theta = \cos \theta(4 \sin \theta + 3) = 0$$

So,

$$\text{Case 1: } \cos \theta = 0 \implies \theta = \frac{\pi}{2}, \frac{3\pi}{2}$$

$$\text{Case 2: } 4 \sin \theta + 3 = 0 \implies \sin \theta = -\frac{3}{4}$$

For $\sin \theta = -\frac{3}{4}$ in $[0, 2\pi)$, $\theta = \arcsin\left(-\frac{3}{4}\right)$ and $\pi + \left|\arcsin\left(-\frac{3}{4}\right)\right|$:

Let $\alpha = \arcsin\left(\frac{3}{4}\right) \approx 0.848$.

So, $\arcsin\left(-\frac{3}{4}\right) \approx -0.848$, so solutions are $\theta_1 = 2\pi - 0.848 \approx 5.435$, $\theta_2 = \pi + 0.848 \approx 3.990$.

Thus,

$$\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \pi + \arcsin\left(\frac{3}{4}\right), 2\pi - \arcsin\left(\frac{3}{4}\right)$$